

Md Monjurul Karim

SIAT, CAS, Shenzhen, China
+86 132 6189 0899 · karim@siat.ac.cn

Interests: Agentic and decentralized AI, blockchain and Web3, Internet of Things (IoT), distributed systems, cloud and edge computing. [Orcid]

PROFESSIONAL EXPERIENCE	Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences Shenzhen, China Postdoctoral Research Fellow and Assistant Researcher Nov. 2023 – Current - Independent and collaborative research on decentralized AI, autonomous AI agents, AI governance, blockchain systems, edge computing, and trustworthy distributed intelligence. - Project design, prototype development, system implementation, experimental validation, and high-quality academic publication. - Interdisciplinary research activities in distributed intelligent systems, trustworthy AI, and networked computing infrastructures. Southern University of Science and Technology Shenzhen, China Research and Teaching Assistant Jul. 2021 – Sep. 2023 - Project <i>Distributed Resource Management for the Integrated Blockchain-Mobile Edge Computing System</i>. - Academic research and teaching activities. Ericsson Information Communication Ltd. Xi'an, China Network Engineer (Internship) Feb. 2015 – Mar. 2015 - Radio Access Network design and optimization.
EDUCATION	Beijing Institute of Technology Beijing, China Ph.D. in Computer Science and Technology Sep. 2016 – Jun. 2023 Thesis: <i>Research on QoS-aware Software-Defined Information-Centric Networking Solutions</i> Advisor: Zhu Liehuang Northwestern Polytechnical University Xi'an, China M.Eng. in Computer Science and Technology Sep. 2012 – Mar. 2015 Thesis: <i>Source Code-based Buffer Overflow Detection</i> Advisor: Yao Ye Northwestern Polytechnical University Xi'an, China B.Eng. in Computer Science and Technology Sep. 2008 – Jul. 2012 Thesis: <i>Filtering packet from a sniffer type application on the IPv6 based networking environment</i> Advisor: Jiang Xuefeng
PROJECTS	Role: Co-PI; <i>AI-Enhanced Explainable Risk Intelligence Platform for Stablecoin Compliance</i>; Hangzhou StablePay Technology Co., Ltd.; Amount: RMB 120,000. Role: Technical Lead; <i>IT Consulting Services for JD Technology's Stablecoin Business</i>; JD Technology Co., Ltd.; Amount: RMB 350,000.
PATENTS	<i>Task Processing Method, Apparatus, System, and Electronic Device</i>. Chinese invention patent, Application No. CN202510684562.0, 2025.
SELECTED PUBLICATIONS	Complete list: Google Scholar · H-Index: 11 · i10-Index: 12 · Citations: 1140+ M. M. Karim, Q. Qu, K. Sharif, M. Muzammal, and S. Biswas, "MTC-SBC: Reputation-based service provision for multi-tier computing-enabled sharded blockchain," <i>Future Generation Computer Systems (FGCS)</i>, vol. 182, p. 108 499, Sep. 1, 2026. M. M. Karim, S. Khan, Q. Qu, M. Muzammal, K. Sharif, and S. Biswas, "From trust to augmentation: A comprehensive survey on synergistic integration of decentralized and generative intelligence," <i>Computer Science Review (COSREV)</i>, vol. 61, p. 100 936, Aug. 1, 2026. D. H. Van, M. M. Karim, and Q. Qu, "Just-in-time historical state reconstruction for low-latency financial trading with large language models," <i>AI</i>, vol. 7, no. 4, p. 117, Apr. 2026. M. M. Karim, D. Hoang Van, and Q. Qu, "Securing decentralized finance: A comprehensive survey of maximal extractable value and its countermeasures," <i>Blockchain: Research and Applications (BCRA)</i>, p. 100 455, Feb. 2, 2026. A. Bilal, K. Sharif, L. Zhu, F. Li, C. Xu, and M. M. Karim, "HySLA: Hybrid DPoS-DAG model for secure, scalable, low-latency access control in Internet of Vehicles," <i>IEEE Internet of Things Journal (IOTJ)</i>, pp. 1–1, 2026.

6. A. Bilal, K. Sharif, L. Zhu, F. Li, C. Xu, and **M. M. Karim**, "Evaluation to integration: Hybrid feature selection framework with ensemble machine learning for intrusion detection," *IEEE Transactions on Dependable and Secure Computing (TDSC)*, pp. 1–16, 2026.
7. D. H. Van, **M. M. Karim**, and Q. Qu, "Probabilistic optimization of top-k aggregation in distributed environments," in *Web and Big Data (APWeb-WAIM 2025)*, Springer Nature, 2026, pp. 566–580.
8. **M. M. Karim**, Q. Qu, Y. Cai, T. Liu, and X. Meng, "Bitcoin reimaged: A comprehensive study of ordinals and inscriptions protocols for Web3 asset innovation," *Blockchain: Research and Applications (BCRA)*, p. 100 379, Sep. 8, 2025.
9. **M. M. Karim**, S. Khan, D. H. Van, X. Liu, C. Wang, and Q. Qu, "Transforming data annotation with AI agents: A review of architectures, reasoning, applications, and impact," *Future Internet*, vol. 17, no. 8, p. 353, Aug. 2025.
10. **M. M. Karim**, D. H. Van, S. Khan, Q. Qu, and Y. Kholodov, "AI agents meet blockchain: A survey on secure and scalable collaboration for multi-agents," *Future Internet*, vol. 17, no. 2, p. 57, Feb. 2025.
11. **M. M. Karim**, K. Sharif, S. Biswas, Z. Latif, Q. Qu, and F. Li, "CIC-SIoT: Clean-slate information-centric software-defined content discovery and distribution for Internet of Things," *IEEE Internet of Things Journal (IOTJ)*, vol. 11, no. 22, pp. 37 140–37 153, Nov. 2024.
12. L. Zhu, **M. M. Karim**, K. Sharif, C. Xu, and F. Li, "Traffic flow optimization for UAVs in multi-layer information-centric software-defined FANET," *IEEE Transactions on Vehicular Technology (TVT)*, vol. 72, no. 2, pp. 2453–2467, Feb. 2023.
13. M. W. A. Ashraf, C. Huang, A. Raza, K. Sharif, **M. M. Karim**, and S. Huang, "Forwarding and caching in video streaming over ICSDN: A clean-slate publish-subscribe approach," *Computer Networks (ComNet)*, vol. 219, p. 109 433, Dec. 24, 2022.
14. W. Ni, S. Zhu, **M. M. Karim**, A. Asheralieva, J. Kang, Z. Xiong, and C. Maple, "Lagrange coded federated learning (L-CoFL) model for Internet of Vehicles," in *2022 IEEE 42nd International Conference on Distributed Computing Systems (ICDCS)*, Jul. 2022, pp. 864–872.
15. Z. Latif, K. Sharif, F. Li, **M. M. Karim**, S. Biswas, M. Shahzad, and S. P. Mohanty, "DOLPHIN: Dynamically optimized and load balanced path for inter-domain SDN communication," *IEEE Transactions on Network and Service Management (TNSM)*, vol. 18, no. 1, pp. 331–346, Mar. 2021.
16. L. Zhu, **M. M. Karim**, K. Sharif, C. Xu, F. Li, X. Du, and M. Guizani, "SDN controllers: A comprehensive analysis and performance evaluation study," *ACM Computing Surveys (CSUR)*, vol. 53, no. 6, 133:1–133:40, Dec. 6, 2020.
17. I. Alam, K. Sharif, F. Li, Z. Latif, **M. M. Karim**, S. Biswas, B. Nour, and Y. Wang, "A survey of network virtualization techniques for Internet of Things using SDN and NFV," *ACM Computing Surveys (CSUR)*, vol. 53, no. 2, 35:1–35:40, Apr. 16, 2020.
18. Z. Latif, K. Sharif, F. Li, **M. M. Karim**, S. Biswas, and Y. Wang, "A comprehensive survey of interface protocols for software defined networks," *Journal of Network and Computer Applications (JNCA)*, vol. 156, p. 102 563, Apr. 15, 2020.

ACADEMIC SERVICE

Conference Organization

- Award Chair, IEEE International Conference on Big Data Science and Engineering
- Technical Program Committee Member, International Conference on Communication Systems and Networks

Peer Review

- All the leading journals and conferences of IEEE, ACM, Elsevier, and Springer.